

AACHEN, 2021

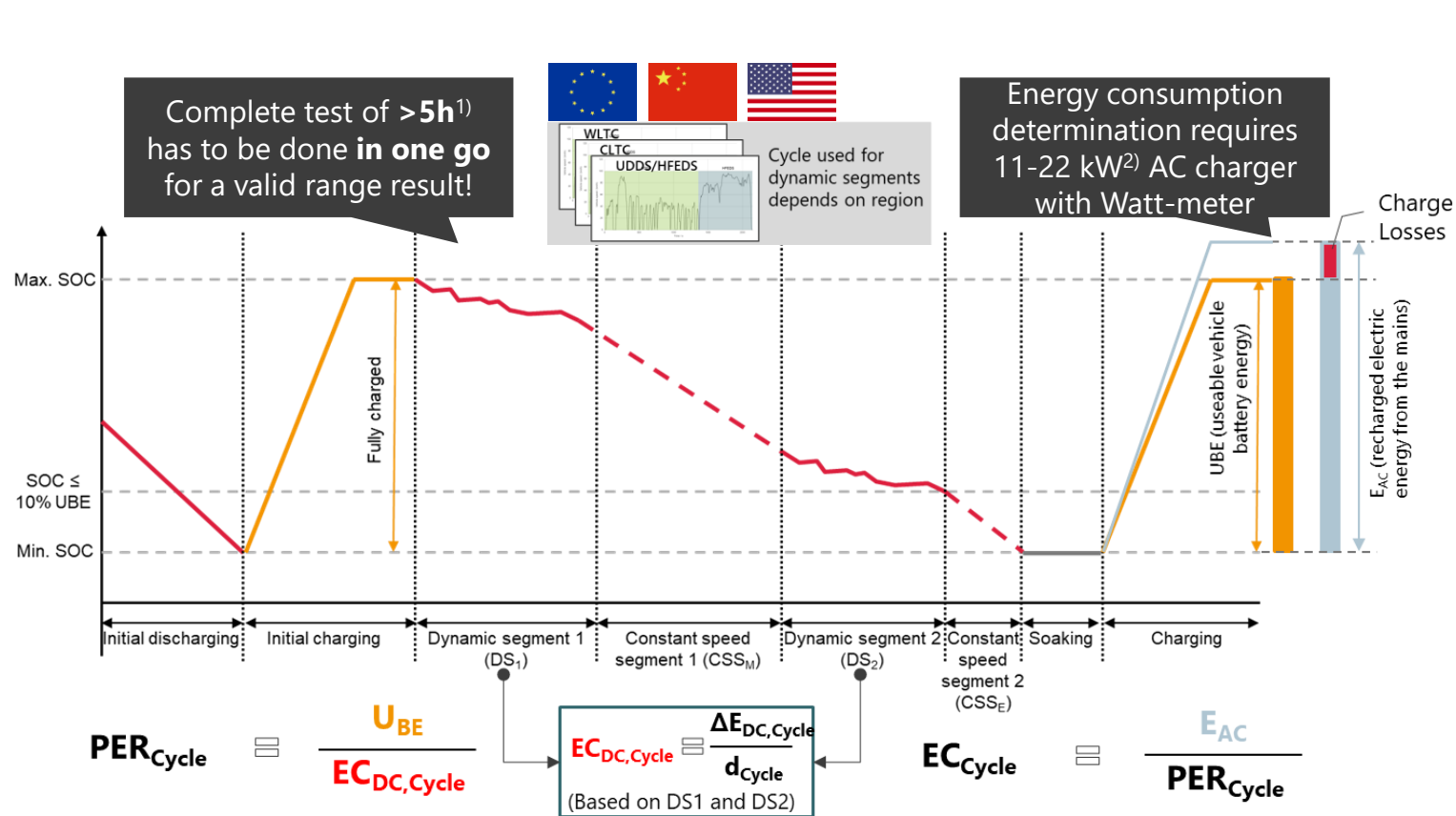
BEV RANGE REGULATIONS WLTP



We can apply different procedures at the chassis dynamometers to comply with different regulations and test requests



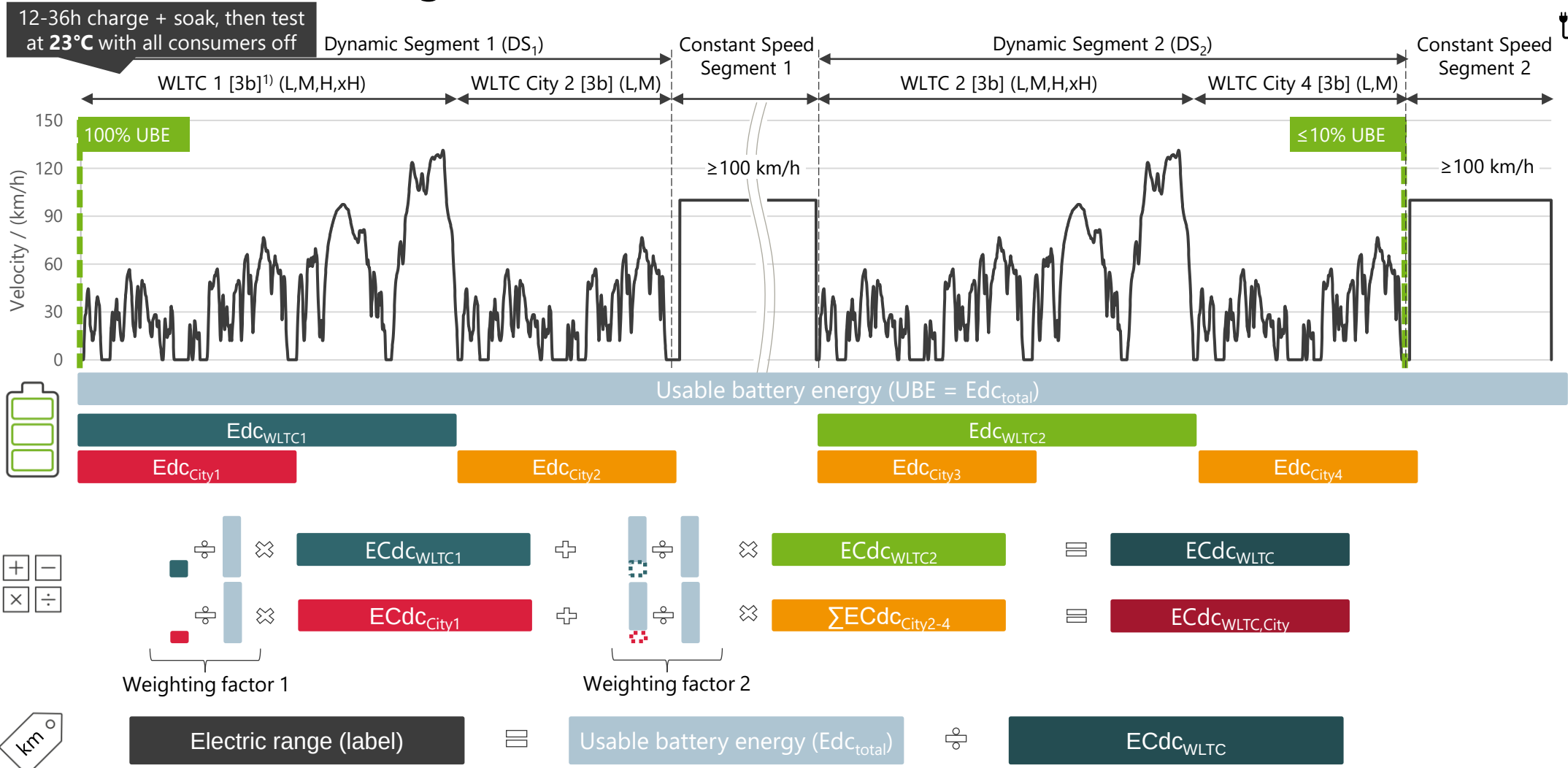
SHORTENED PROCEDURE SCHEMATIC APPLIED BY EU WLTP, US FTP AND CHINA CLTC



- Similar pattern for BEV shortened procedure is applied in Europe, Japan (both WLTP) and USA, (FTP)
 - China introduced new procedure from 10/2021 with same approach using country specific CLTC-cycle for Dynamic Segment
 - India is expected to apply WLTP standard in upcoming years as well
- Country specific application and calculation pattern is explained on following slides

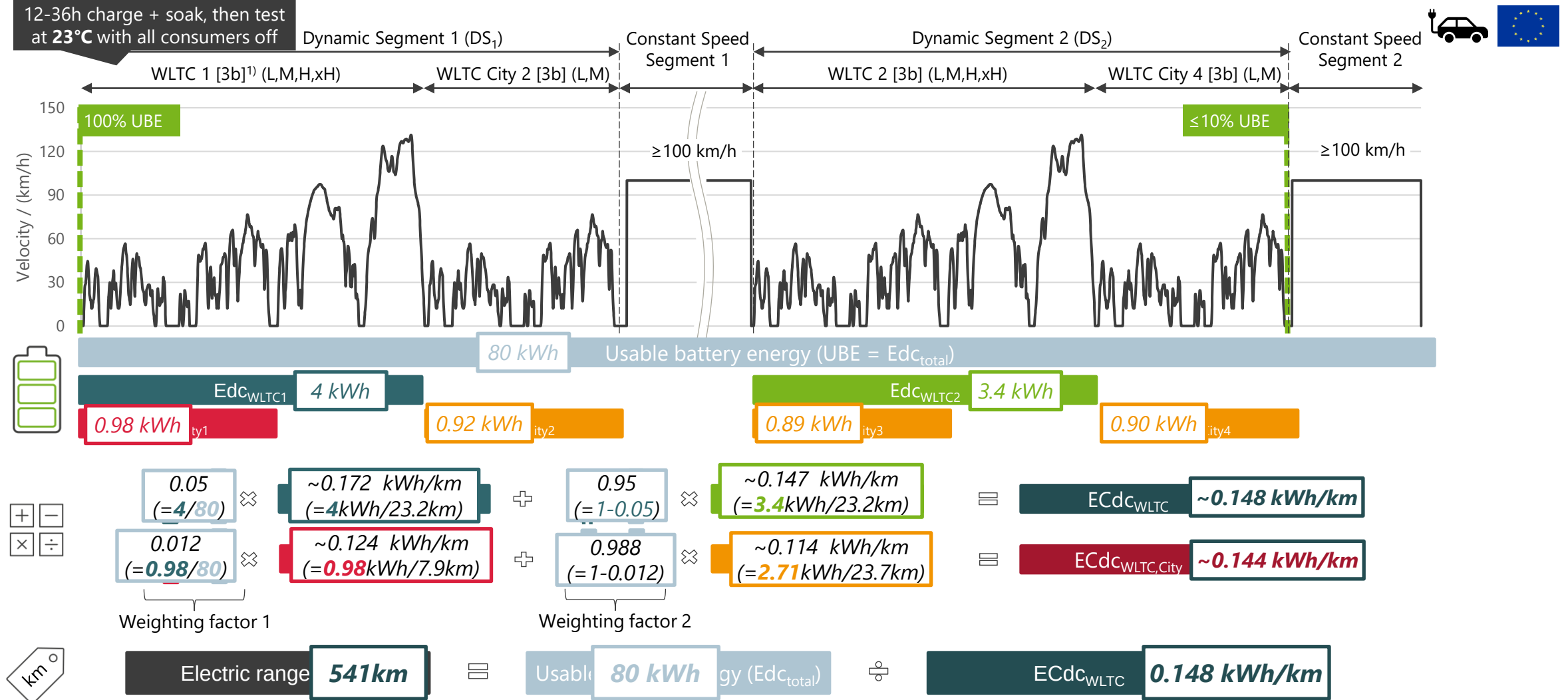
1) Typical duration for BEV with 60 kWh battery energy 2) Current regulations require 11 kW, but existing draft for DIN 70800 already requires 22 kW
 UBE=Usable Battery Energy (over complete procedure), E = Energy measured at battery clamps, EC = Energy Consumption per distance,
 DC = Direct Current (w/o charging losses), AC = Alternating Current (incl. charging losses), d = distance, PER = Pure Electric Range, SOC = Battery State of Charge
 Sources: EU WLTP Regulation (1151/2017), US FTP Regulation (SAE J1634, CFR 40 Parts 86,600), China CLTC (GB/T 18686.1), DIN 70080

WLTP shortened procedure (STP) and weighting factors to determine label values for electric range



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 1) All electrified vehicles are defined as "class 3" vehicles independent of power to weight ratio and must apply WLTC class 3b (downscaling options for low power/speed vehicles described here)
 Source: EU Regulation 1151/2017 amended by EU Reg. 49/2020 as of 21.01.2020

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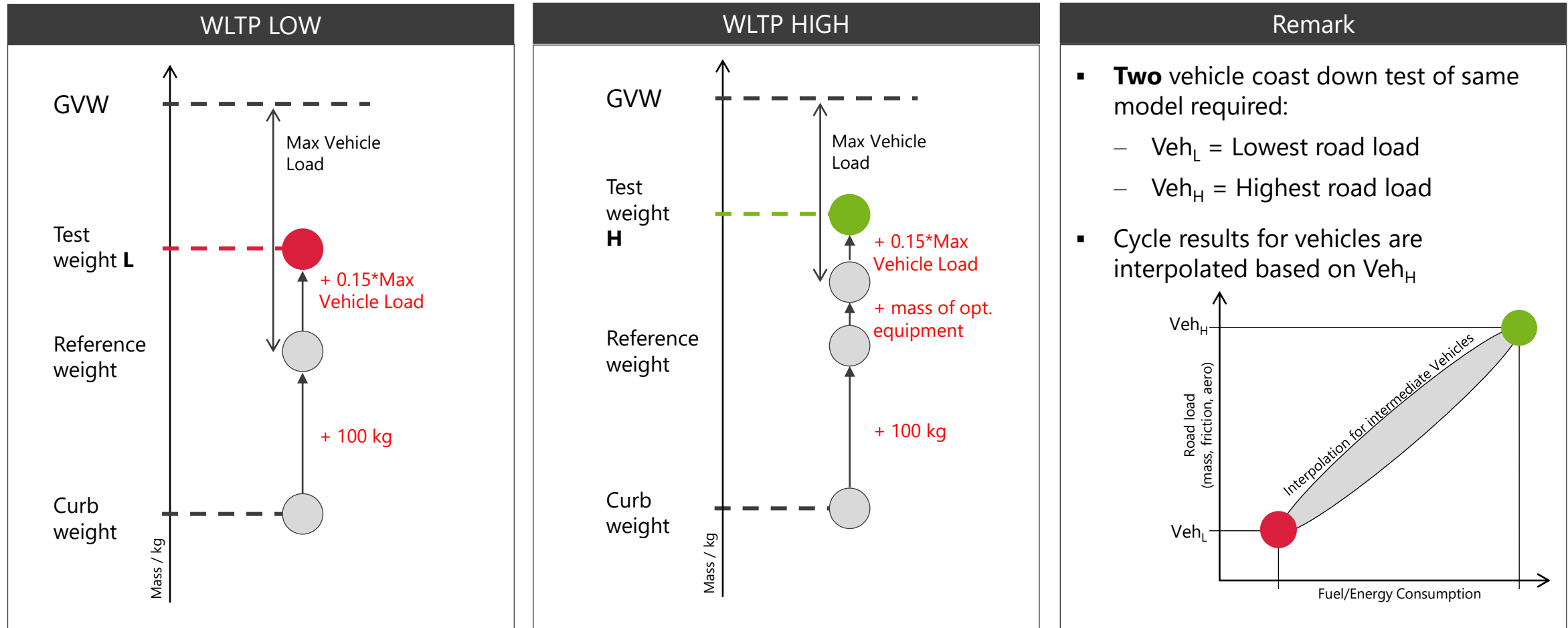
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EU WLTP test mass definition for energy consumption and range



TEST MASS FOR WLTP

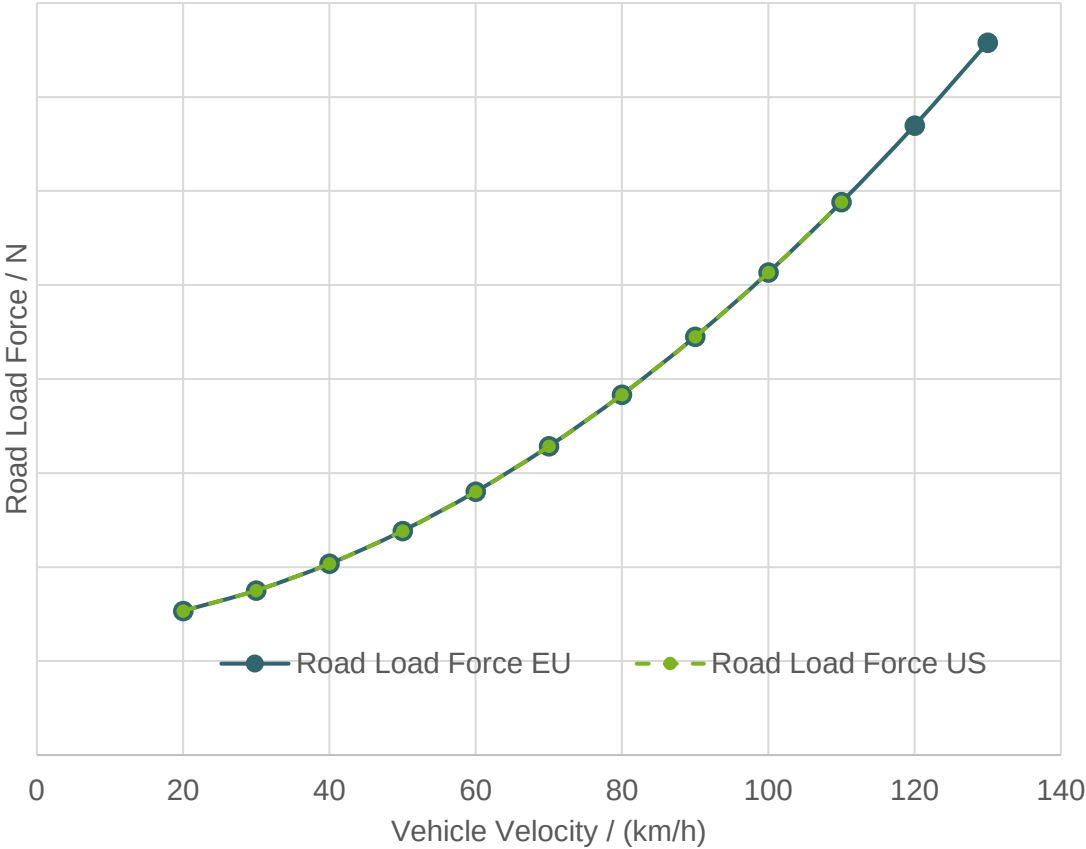


Road Load Derivation Settings differ significantly between WLTP EU and US FTP application!



KEY DIFFERENCES AND RECOMMENDATIONS

DYNO REFERENCE POINTS



KEY SETTINGS

	WLTP EU 
Velocity Band	15- 135 km/h
Reference Points	20, 30, ... , 130 km/h
Velocity Coast Down Start	140-145 km/h (+10-15 above max. ref.)
Recommended RLD mode	Iterative Mode
Acceleration Mode (for iterative RLD)	Driver acceleration only
Force Tolerance	+/- 10N
Number of verification runs	1 Run (total of 2 consecutive Runs)
Maximum runs	
Warm-Up Cycle	1x WLTC
Initial Set Coefficients (Also to be used for Warm-Up Cycle!)	2WD&4WD K0=0.5*F0 K1=0.2*F1 K2=1.0*F2